Semantic Segmentation of Large-Size VHR Remote Sensing Images Using a Two-Stage Multiscale Training Architecture

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***Abstract*— Very-high resolution (VHR) remote sensing images (RSIs) have significantly larger spatial size compared to typical natural images used in computer vision applications. Therefore, it is computationally unaffordable to train and test classifiers on these images at a full-size scale. Commonly used methodologies for semantic segmentation of RSIs perform training and prediction on cropped image patches. Thus, they have the limitation of failing to incorporate enough context information. In order to better exploit the correlations between ground objects, we propose a deep architecture with a two-stage multiscale training strategy that is tailored to the semantic segmentation of large-size VHR RSIs. In the first stage of the training strategy, a semantic embedding network is designed to learn high-level features from downscaled images covering a large area. In the second training stage, a local feature extraction network is designed to introduce low-level information from cropped image patches. The resulting training strategy is able to fuse complementary information learned from multiple levels to make predictions. Experimental results on two data sets show that it outperforms local-patch-based training models in terms of both accuracy and stability.**

***Index Terms*— Convolutional neural network, deep learning, remote sensing, semantic segmentation.**

I. Introduction

With the rapid progress of Earth observation technologies, a large volume of very high resolution (VHR) remote sensing images (RSIs) is available. Semantic segmentation of VHR RSIs, which refers to the per-pixel classification or labeling of images [1], is a crucial step for the interpretation of ground information. Since deep convolutional neural networks (DCNNs) have been greatly developed in processing natural images [2]–[7], they have been extended to the semantic segmentation of VHR RSIs in recent years. However, there are significant differences between natural images and VHR RSIs, among which spatial size is a nonnegligible factor. VHR RSIs have usually a much larger size than typical natural images used in computer vision applications. For example, images in the PASCAL VOC data set [8] (an open benchmark for semantic segmentation of natural images) have a size of 500 × 375 pixels, whereas images in the ISPRS semantic labeling data set (Potsdam area) [9] have 6000 × 6000 pixels and the original images in the Beijing land-use classification data set have 47 244 × 47 244 pixels.

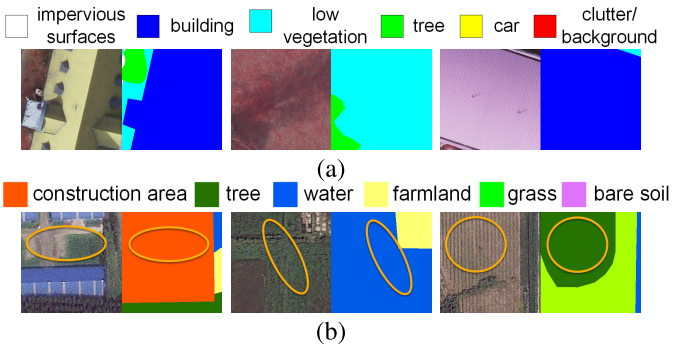


Fig. 1. Examples of pairs of cropped RSIs and their reference maps. Examples from the (a) ISPRS Potsdam data set. The cropped image patches cover only incomplete parts of the objects and (b) Beijing land-use data set. There are discrepancies between the image information and the reference maps (highlighted with yellow circles).

The significant difference in the spatial size makes it impossible to give a whole VHR RSI as input to DCNN models; otherwise, the computational costs would be unaffordable [10]. Most of the existing studies use cropped RSIs as input and stitch the output images as final predictions [11], [12]. This strategy has the limitation of failing to make use of context information at a higher level of abstraction. In Fig. 1, we provide some examples of cropped RSIs with a size of 320 × 320 pixels. One can observe in Fig. 1(a) that the cropped local patches from the ISPRS data set only cover part of the ground objects. Thus, these cropped images can provide only spectral and texture information of a part of the ground objects, whereas their context information is incomplete or missing. Moreover, in practical applications, there are usually large areas of discrepancies between images and the reference maps due to coarse manual labeling or collection of reference data and images at different times. In Fig. 1(b), some examples of the discrepancies between images and the corresponding reference maps (from the Beijing land-use data set) are given. In these cases, a large proportion of areas in cropped image

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patches are providing wrong training information, which has a negative impact on learning an effective model.

In this article, we present an approach that better exploits the global context for an accurate and stable semantic segmentation (classification) of large-scale RSIs. We propose a two-stage multiscale training architecture. The two stages are focused on the extraction of high-level information and local information, respectively. The main contributions of this article are as follows.

1) We propose to use downscaled RSIs as input for the extraction of high-level features, so that the context information can be learned from a global level and the training robustness is greatly enhanced.
2) We design a multilevel DCNN architecture to extract local features and recover spatial resolution from cropped local image patches.
3) We propose a feature fusion module to incorporate the information learned from each separate training stage (i.e., semantic information learned from downscaled images and the local details learned from cropped image patches). This complementary information is incorporated to yield dense predictions.

We evaluated the proposed approach on two VHR RSI data sets: 1) the ISPRS Potsdam data set, which is a publicly available benchmark data set for 2-D semantic labeling and 2) the Beijing land-use classification data set, which is a large-scale VHR satellite image data sets.

The remainder of this article is organized as follows. Section II reviews the related work. Section III gives a detailed description of the proposed approach. Section IV illustrates the data sets used in the experiments and the performance assessments of the methods. Section V analyses the experimental results. Section VI draws the conclusions of this article and discusses future research directions.

II. Related Work

Most of the recently developed approaches to the task of semantic segmentation are based on fully convolutional networks (FCNs) [5], a family of neural networks that show high computational efficiency for dense prediction and end-to-end learning. Although FCNs have a hierarchical architecture that can effectively embed high-level semantic information into feature maps, there is the problem of recovering detailed localization information to generate an accurate segmentation map. In this section, first, we review the development of strategies for both semantic embedding and localization, then we briefly discuss the use of CNNs in semantic segmentation of RSIs.

A. Semantic Segmentation of Natural Images

1) Encoder–Decoder Strategy: To perform the task of semantic segmentation, the output of classification models must be recovered to the same size as the input. To this end, an unpooling and deconvolution operation was introduced in [13], which learns an upsampling process to generate dense and precise object masks. Since then the encoder–decoder strategy has been widely used. A typical encoder–decoder network contains an encoder module that gradually reduces the spatial size of feature maps and captures higher semantic information, and a decoder module that gradually recovers the spatial information. SegNet [14] and U-Net [15] are typical networks using this architecture. They both contain a contracting path to capture context and a symmetric expanding path that enables precise localization (the latter is particularly suited for the segmentation of large-scale medical images). RefineNet further develops this architecture into a multipath network and uses ResNet as its encoding module [10].

2) Dilated/Astrous Convolution: The size of receptive field is known to be crucial for embedding high-level features. In order to enlarge the receptive field while using less pooling layers (which causes a loss of spatial information), dilated convolution (or the so-called atrous convolution) has been proposed in [16]. The dilated/atrous convolution operators can apply the same filter at different ranges using different dilation factors, thus systematically aggregating multiscale contextual information without losing spatial resolution. This module has been adopted and improved in DeepLab V1–V3 [17], [18], [23] and combined with the ResNet [6] as backbone in Dilated Residual Network [19]. Although successful, dilated/atrous convolution has the limitation to be computationally expensive and yielding gridding effects.

3) Spatial Pyramid Pooling: Spatial pyramid pooling (SPP) was proposed as an extension of the Bag-of-Words model [20]. It partitions the image into divisions from finer to coarser levels and aggregates local features in them. This model has been incorporated with a CNN and thus named SPP-net to exploit multiscale descriptors for various objects [21]. PSPNet extends this study by using a pyramid parsing module to acquire different subregion representations [22]. DeepLab proposes an atrous SPP (ASPP) by implementing pyramid pooling with atrous convolution [23].

To conclude, the balance between powerful feature representation and preservation of spatial details is a dilemma for the task of semantic segmentation. All these literature works are trying to enlarge the receptive field of the networks with less possible loss of spatial information. Additionally, weakly supervised learning [24] and the use of adversarial learning for domain adaptation [25] are also promising strategies to alleviate the data-hungry problem of DCNNs.

B. Semantic Segmentation of RSIs

Semantic segmentation of RSIs can be used in various applications such as change detection, extraction of ground objects, and generation of thematic maps. With their success in computer vision, CNNs quickly drew research interest in the field of remote sensing. The use of CNNs resulted in a breakthrough in semantic segmentation of RSIs.

Due to the scarcity of training data, researches on semantic segmentation for RSIs began years after the research on natural images. An early research was given on the generalization performance of pretrained CNNs applied to RSIs [26]. In [27], a multitask joint sparse representation is proposed to deal with the high interband correlation problem of hyperspectral RSIs. Several studies incorporate multiple models to increase

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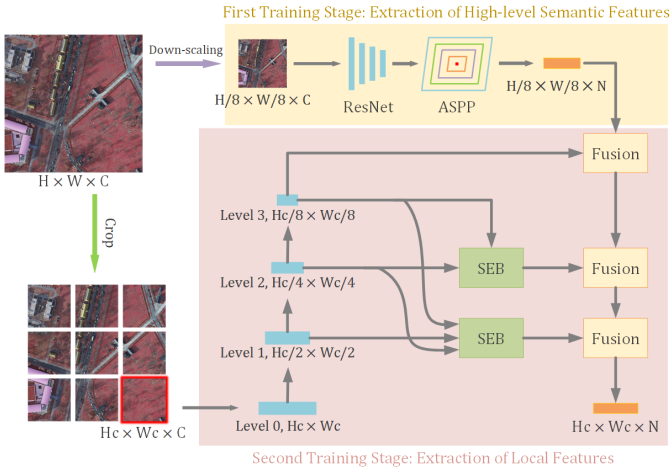


Fig. 2. Overview of the proposed two-stage multiscale training architecture for the semantic segmentation of large-scale RSI. H, W, and C: height, width, and dimensions of the input image, respectively. H_C and W_C: height and width of the cropping window, respectively. N: number of classes.

the prediction certainty. In [28], CNN and crafted features were fused according to the pixel values on their probability maps to yield dense prediction. A hybrid Markov random field (MRF)-CNN model was proposed in [29] using a fusion strategy that quantitatively calculate the uncertainty distribution of CNN outputs. PSPNet and ResNet were incorporated in [30] for the segmentation of VHR RSIs. To enhance the precision on predicting object boundaries, an edge loss to enhance the preservation of object boundaries was introduced in [12]. Another study [1] added edge detection in SegNet to make class boundaries explicit. Object level segmentation has also been investigated to cope with the intraclass variability problem for classifying VHR RSIs [31]–[33]. To solve the problem of scale variation of RSIs, a multiscale training strategy was proposed in [34]. Numerous studies gave efforts on exploiting potential information from other types of data collected in the same region as RSIs to improve the segmentation performance. In [35], satellite images and the corresponding OpenStreetMap (OSM) data were used in a pretraining phase to alleviate the data-hungry problem of CNNs. OSM data have also been used as an input to SegNet in [36], as well as Lidar and multispectral data in [37]. A multiple-task feature fusion network integrating RGB bands, digital surface model (DSM), and an infrared (IR) band using U-Net has shown better performance compared with any single type of input data [38]. DSM was also used in [39] as an individual backend to remove false predictions.

From a general perspective, existing works for semantic segmentation on RSIs mostly employ multiple models to get integrated results or fuse different types of data to better exploit the ground information. At the moment, to the best of our knowledge, there are no existing works giving importance to the specific properties of RSIs related to their large size.

III. Proposed Approach

In this article, we propose a two-stage multiscale training architecture that is specially tailored for the semantic segmentation of large-scale VHR RSIs. The proposed architecture



(a) (b)

Fig. 3. Visualization of the VRF of a pixel. (a) VRF on an image patch. (b) VRF on a downscaled image patch.

is shown in Fig. 2. It contains two separate training stages. In the first training stage, downscaled RSIs are as input to the network to enable the model to learn high-level information from a wider spatial range. The second stage of training is focused on extracting local information from the cropped image patches by taking into account the features learned from the higher levels as semantic supervision. As a result, the proposed architecture is able to incorporate complementary information learned from multiple levels to make the final predictions of the semantic maps.

A. First Training Stage: Extraction of High-Level Semantic Features

As mentioned, we use downscaled RSIs as input to the first training stage. There are two main motivations for this stage.

1) *Enlarging the Valid Receptive Field (VRF) of the Network:* VRF is known to be crucial for the classification of image objects [40], [41]. If the VRF is large enough to hold the whole object and its surroundings, a DCNN architecture has a better chance to learn potential representations of their correlations. However, most works on semantic segmentation of RSIs use cropped images as input. Their cropping window size is usually 320 × 320, 256 × 256 or lower. Consequently, the VRF of the employed models is far from covering the entire object, especially for the semantic segmentation of VHR RSIs. Fig. 3(a) shows an example of the VRF on a cropped image patch using a DCNN architecture. In this case, only texture and spectral information can be collected from the given image patch. As a comparison, Fig. 3(b) shows the VRF on a downscaled image patch. The use of downscaled images enables the model to learn more context information.
2) *Increasing Training Stability:* As shown in Fig. 1(b), there are often some discrepancies between original images and the related reference data used for the training of the network (mislabeled training data). Using downscaled images as input equals using a much larger training batch size. Since the downscaled patches cover a larger image area, the impact of mislabeled areas is less severe. Therefore, the training becomes more stable and has a better chance to converge to an effective model.

In this training stage, we adopt a DCNN architecture that is similar to DeepLabv3+ [42]. It contains a modified version of ResNet101 as a backbone network for feature embedding. The number of channels of its first convolutional layer is changed

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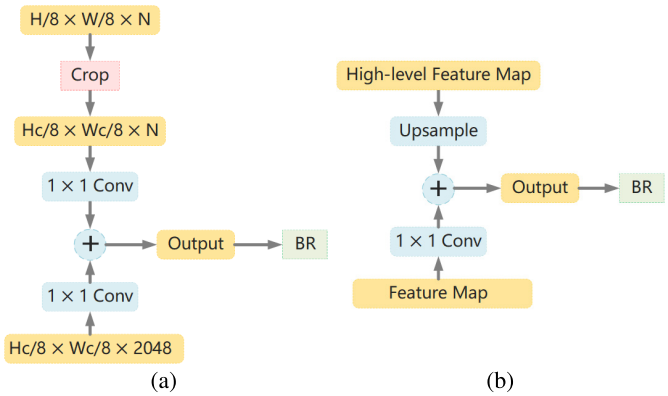


Fig. 4. Structure of the feature fusion blocks in Fig. 2 ("⊕" denotes pixel-wise sum). (a) Feature fusion with the output of the first training stage. (b) Feature fusion with high-level features.

according to the input data and a set of dilated kernels with a different rate (1, 2, and 5) are added to its last "bottleneck" unit. On top of the backbone, the ASPP module is introduced to better exploit information from the surrounding pixels. This is a structure designed with multiple parallel filters with different rates to collect information from multiple scales. The output is further enlarged and concatenated with the features from the lower levels to recover the spatial size. Finally, this network forward an output with N channels and has the same spatial size as the input downsampled data, where N refers to the number of object categories.

B. Second Training Stage: Extraction of Local Features

In the second training stage, we use cropped image patches at their original resolutions to learn detailed local information. Since this training stage involves only the extraction of low-level information such as texture and spectral information, complex structures such as ASPP are not adopted in order to reduce the computational cost. To recover spatial resolution with the guide of high-level information, we designed a multilevel architecture to learn coarse-to-fine grained features. The connection between different layers is established by the feature fusion blocks. Likewise, there is also a similar block on the highest level of the model to introduce the features learned during the first training stage. Inspired by the work in [43], a semantic embedding branch (SEB) is also used to improve the feature fusion between different layers.

1) Feature Fusion Block: As illustrated in Fig. 4, we designed two versions of the feature fusion block. The structure in Fig. 4(a) is designed to introduce the high-level features learned from the first training stage. These features are first cropped to the same regions related to the local image patches and then added on top of the local feature extraction network to provide semantic guidance from the higher levels. Meanwhile, the structure in Fig. 4(b) is responsible for feature fusion between different levels. Features learned from higher levels are upsampled and added to the current feature layer. There is also a boundary refinement (BR) module introduced at the end of each feature fusion block to enhance the boundary fusion. It consists of a residual convolutional unit, details of which can be found in [40].

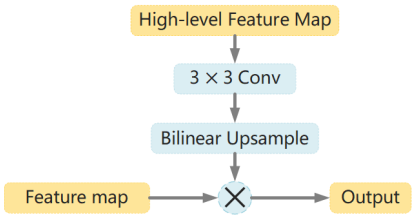


Fig. 5. Structure of the SEB in Fig. 2. If there are more than one group of semantic features from high levels, they will be upsampled to the same resolution ("⊗" denotes pixel-wise multiplication).

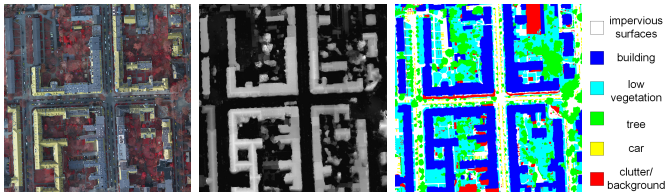


Fig. 6. Example of an area in the ISPRS Potsdam data set. (a) Training image (IRRG). (b) nDSM. (c) Reference map.

2) SEB: The overall idea of the SEB is to include more semantic information from higher levels to improve the efficiency of the feature fusion. Indeed, as discussed in [43], the use of only low-level features with little semantic information is insufficient to recover the semantic resolution. The details of SEB are illustrated in Fig. 5. This structure enables resolution recovery of feature map by including multiple features from higher abstraction layers.

IV. Description of Data Sets and Design of Experiments

In this section, we test the proposed two-stage multiscale training architecture on two VHR RSI data sets. We give descriptions of the tested data sets, implementation details, and evaluation metrics, respectively, in Sections III-A–III-C.

A. Description of Data Sets

1) ISPRS Potsdam Data set: This is a well-known data set provided for the 2-D labeling contest [9]. It contains 38 image tiles together with the corresponding normalized DSM (nDSM) data. Each tile has 6000 × 6000 pixels and 4 channels: red (R), green (G), blue (B), and near IR. The ground sampling distance (GSD) is up to 5 cm. Following the practice in [44], we use 24 tiles for training and the remaining 14 tiles for testing. The reference data are labeled according to the following categories of ground objects: impervious surfaces, building, low vegetation, tree, car, and clutter/background. An example of the training data and the corresponding reference map in this data set is given in Fig. 6. It should be mentioned that in some studies [1], [29] the clutter/background class is omitted and not predicted due to its relatively small proportion in the total number of pixels, whereas some other studies included this class into account [30], [44], which may cause confusion in performance evaluation. In this article, we take this category into both prediction and performance assessment.

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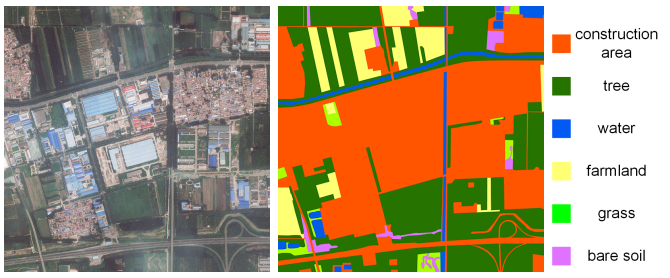


Fig. 7. Sample area of the Beijing land-use data set. (a) Training image. (b) Reference map.

2) Beijing Land-Use Data Set: This is a satellite data set collected by the Beijing-2 constellation in June 2018. It contains 4 image tiles, each has 47 244 × 47 244 pixels. The collected images are composed of three optical bands: red (R), green (G), and blue (B). The GSD is 0.8 m. Since the original size of images is beyond the limit of our computational resources, the 4 tiles are split into 196 images (49 images for each original tile); 147 images are selected for training, the remaining 49 are used for testing. Each training and testing image has 6720 × 6720 pixels, which is still large enough to cover sufficient context information. The reference map is annotated into six land-use categories: construction area, tree, water, farmland, grass, and bare soil. An example of the training data and the corresponding reference map in this data set is given in Fig. 7. Note that, since the reference map is originally annotated with attention driven by the application instead of semantic segmentation, it is not labeled in a pixel-wise manner (i.e., it is coarse-grained and has significant discrepancies with the ground truth). Examples of the discrepancies between images and their corresponding reference maps are shown in Fig. 1(b). This circumstance can be a good test on semantic segmentation methods in terms of robustness and tolerance to mislabeled data, since coarse-grained reference data are quite common in many practical applications. The reference maps for the testing set are relabeled based on the ground truth information to ensure an accurate assessment of the effectiveness of the studied approaches.

B. Implementation Details

The proposed model is implemented on the PyTorch deep learning platform [45]. All experiments are conducted on a server with an Intel Xeon CPU E5-1650 v2 CPU, 16 GB of memory (RAM), and an Nvidia Quadro P6000 graphical processing unit (GPU) (24 GB). During the first training stage, the input images are downscaled to 1/8 of their original spatial size. This downsampling rate is set to produce inputs with the same resolution as the feature maps produced by the ResNet backbone in the second training stage. The downsampling method for RSIs is area interpolation (OpenCV implementation) to produce moire-free results. However, for the downsampling of reference maps, nearest interpolation is the only option to keep the labels in "one-hot" form (e.g., no intermediate values can be interpolated between different label values). Data normalization is performed on each spectral channel. To avoid the over-fitting problem, data augmentation is used, including flipping and image rotation. Flipping is performed randomly in the vertical or horizontal direction while loading each batch of data. Image rotation, which takes more calculation time, is performed before the training to generate a set of rotated images. After that, a sliding crop operation is conducted on both original and rotated images. The sliding step is calculated automatically according to the cropping window and the image size, ensuring the whole image is covered with a minimum overlapping area. Fig. 8 illustrates our image rotation and sliding crop method. The second training stage takes much more calculation resources to process large image tiles at their original resolution. Therefore, image rotation is not implemented to reduce the processing and training time, whereas random flipping is still performed. During the second training stage, instead of doing a whole-image cropping, a random cropping operation is conducted at each epoch, which means the cropping window locates randomly over the image. In this way, the effective training area remains the same whereas the training time reduces tremendously. All the backbones involved in our architecture are used with pretrained weights. The ISPRS Potsdam input data are concatenated into five channels (RGBIR and nDSM). Since the pretrained weights of ResNet have only 3 input channels, we copied the weights of two channels (G and R) to the additional IR and nDSM channels. This is an empirical choice reasonable for the initialization of the network. We start the training from a pretty high learning rate (lr = 0.01) and used Adam as the optimization algorithm for gradient descent [46].

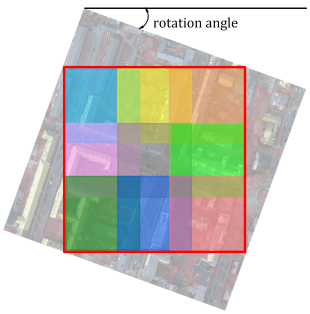


Fig. 8. Image rotation and sliding crop. Red rectangle: calculated maximum cropping area within a rotated image. Colored squares: calculated cropping windows.

C. Evaluation Metrics

We use the most frequently used evaluation metrics in semantic segmentation of RSIs: overall accuracy (OA) and F1 score [1], [30], [47]. OA is the total number of correctly labeled pixels divided by the total number of pixels. The F1 score for a certain class is defined as the harmonic mean of precision and recall

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{1}$$

Here, precision and recall are calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

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TABLE I

RESULTS ON THE ISPRS POTSDAM DATA SET. PER-CLASS F1 SCORE, AVERAGE F1 SCORE (AVG.F1) AND OA ARE LISTED (%). (IMP.SURF.: IMPERVIOUS SURFACE; LOW VEG.: LOW VEGETATION)

Method	Per-class F1 Score					avg.F1	OA
	Imp.Surf.	Building	low veg.	Tree	Car		
UNet	88.00	93.52	77.76	73.43	78.18	82.18	82.81
PSPNet	87.43	93.20	79.55	72.87	78.18	82.25	83.02
SegNet	87.48	96.19	83.24	85.69	88.82	88.28	86.85
FCN	91.98	96.59	85.66	86.86	95.05	91.23	89.58
DeepLabv3+	91.70	96.71	86.68	87.04	93.64	91.15	89.75
Proposed	92.91	97.13	87.03	87.26	95.16	**91.90**	**90.64**

where TP stands for true-positive (namely the number of pixels of a given class that get correctly classified), FP stands for false-positive (indicating the number of pixels of other categories that are wrongly classified into a specific category), and FN represents false-negative (indicating the number of pixels of a given class that are wrongly classified as other categories).

V. EXPERIMENTAL RESULTS

A. Results

We conducted our experiments by comparing the proposed architecture with several classic DCNN architectures, including FCN [5], UNet [15], SegNet [14], PSPNet [22], and the most recent DeepLabv3+ [42]. All these networks are well-known and frequently used in remote sensing literature [12], [39], [44]. PSPNet, FCN and DeepLabv3+ are implemented using resnet101 [6] as the baseline and SegNet uses vgg19 [3] as the backbone. Table I reports the results achieved on the ISPRS Potsdam data set. One can observe that all the models with deep backbones have achieved a relatively higher performance compared with simple architectures such as UNet. To our surprise, the performance of the most simple model FCN is at the same level as that of the state-of-the-art model DeepLabV3+. This will be further discussed in Section V-B. The proposed approach shows an advantage in both the average F1 score and OA. An example of the testing result in one of the areas is shown in Fig. 9. One can observe that, compared with other approaches, the prediction map of the proposed approach has less broken segments and the boundary of classified objects are more smooth and complete.

Table II reports the experimental results obtained on the Beijing land-use data set. Due to the inaccurate labeling of reference data (as mentioned in Section IV), the tested models are trained with a significant amount of misleading information. The performance of most models is far from being satisfactory, especially for the categories with less training area (such as grass and bare soil). However, our training strategy successfully converges on this data set and achieves the best results in every measurement. The F1 scores for classes grass and bare soil are still poor but significantly better than all other approaches. Fig. 10 shows the prediction maps generated by the tested methods. Although the categories with enough training samples, such as constructions and trees are effectively recognized by the compared approaches, the weak categories such as grass, water, and bare soil are completely neglected by most of them. Only the proposed approach made reasonable predictions for these categories, even if with absolute accuracy to be still significantly impacted.

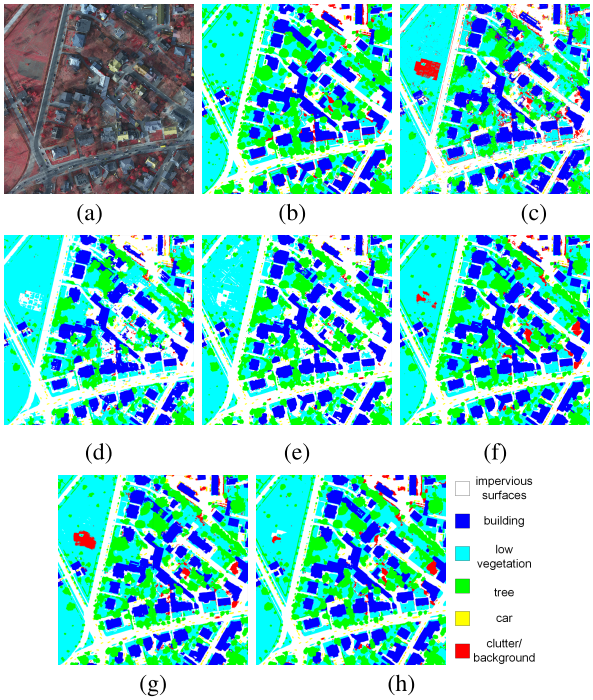


Fig. 9. Classification maps produced by the compared approaches on the ISPRS Potsdam data set. (a) Testing image (IRRG), (b) reference map, (c) UNet, (d) PSPNet, (e) SegNet, (f) FCN, (g) DeepLabv3+, (h) proposed.

TABLE II

RESULTS ON THE BEIJING LAND-USE DATA SET (CONS.: CONSTRUCTION AREA)

Method	Per-class F1 Score						avg.F1	OA
	Cons.	Tree	Water	Farmland	Grass	Bare Soil		
UNet	83.45	74.36	0.00	9.23	8.62	0.00	29.28	66.67
PSPNet	86.60	80.62	0.00	35.76	0.00	0.00	33.83	70.47
SegNet	85.39	80.40	20.32	41.97	0.00	0.00	38.01	69.72
FCN	81.44	79.26	22.93	30.10	7.74	5.56	37.84	68.11
DeepLabv3+	85.20	73.59	0.00	0.00	0.00	0.00	26.46	67.60
Proposed	90.73	86.25	73.72	71.75	31.06	30.07	**63.93**	**82.71**

To evaluate the efficiency of the proposed approach, a comparison of training time is made with FCN and DeepLabv3+. All the compared approaches use ResNet101 as a backbone and have the same settings (batch size, training epochs, and so on). The random cropping operation mentioned in Section IV-B is used for training FCN, DeepLabv3+, and the second stage of our approach to reduce the computational time. The training time taken by different methods are given in Table III. The very basic FCN architecture takes the least time for training, whereas the more complicated DeepLabV3+ costs more time. Our training strategy also uses ASPP in the first training stage, but the downsampling operation to input images significantly reduces the training time. The second training stage of the proposed approach takes slightly more time than FCN. Compared to DeepLabV3+, the total training time of our approach is smaller for the Potsdam data set while higher for the Beijing data set.

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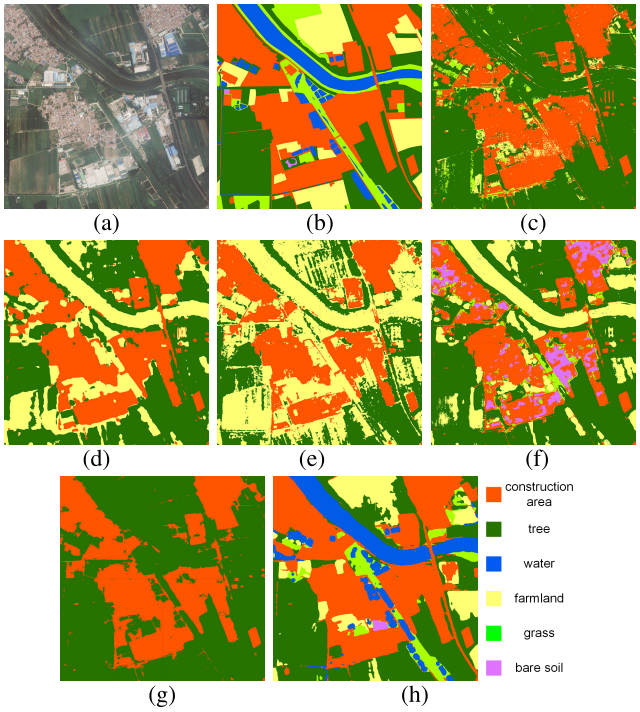


Fig. 10. Classification maps produced by the compared approaches on the Beijing land-use data set. (a) Testing image, (b) reference map, (c) UNet, (d) PSPNet, (e) SegNet, (f) FCN, (g) DeepLabv3+, (h) proposed.

TABLE III
COMPARISON OF TOTAL TRAINING TIME FOR 50 EPOCHS (MINUTES)

Dataset	FCN	DeepLabv3+	Proposed Approach		
			1st stage	2nd stage	Total
Potsdam	38.4	146.4	57.4	42.2	99.3
Beijing	192.0	268.2	90.6	201.6	292.2

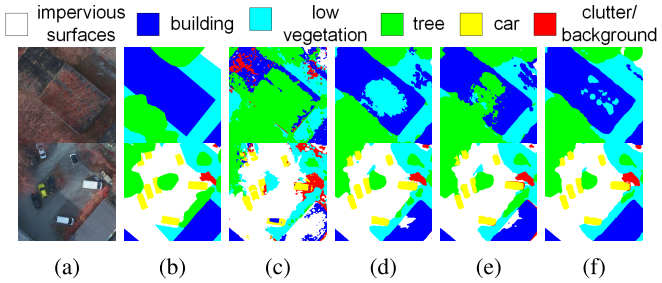


Fig. 11. Comparison of classification maps on image patches (ISPRS Potsdam data set). (a) Image patch, (b) reference map, (c) UNet, (d) FCN, (e) DeepLabv3+, (f) proposed.

B. Analysis of Results: Effect of Downscaled Training

As introduced in Section III, the size of the VRF is crucial for encoding context information. Fig. 11 shows a comparison of the predictions on local patches on the ISPRS Potsdam data set. Our experimental results show that shallow models such as UNet tend to produce more fragmented maps. As shown in Fig. 11(c), the predicted object boundaries of UNet are noisy and disconnected. This is because their relatively small VRFs restrict the models from considering the surrounding pixels. In deep models such as SegNet and FCN with ResNet backbones, this phenomenon is alleviated since they have much larger VRFs. DeepLabV3+ was expected to achieve a much better result than FCN due to its ASPP module that can greatly enlarge the VRF. However, due to the limitation of cropping size, its VRF field is also limited to local patches, where no enough context information is included. As a result, the performance of FCN and DeepLabv3+ are at the same level. The proposed approach, due to the use of downscaled images in the first training stage, greatly enlarges the VRF. As shown in Fig. 11(f), the object boundaries in our predictions are more smooth and reliable.

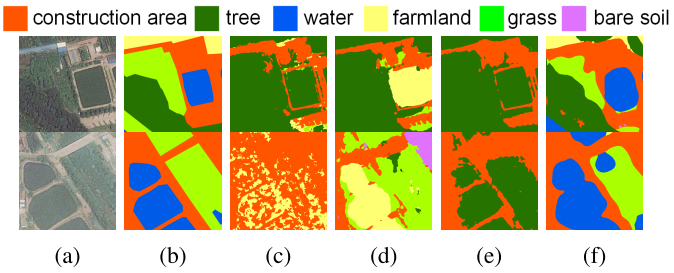


Fig. 12. Comparison of classification maps on image patches (Beijing land-use data set). (a) Image patch, (b) reference map, (c) UNet, (d) FCN, (e) DeepLabv3+, (f) proposed.

The experimental results on the Beijing land-use data set also prove that using downscaled images can effectively improve the training robustness. As shown in Fig. 12, our approach is the only one that makes predictions for all categories. All other approaches neglected several categories, which is a sign of underfitting. This insufficient training of certain categories is caused by the misleading information presented in the training data. However, with the introduction of the additional downscale training stage, our approach is more robust to misleading annotations and the prediction results are closer to the ground truth.

To quantitatively evaluate the effect of using downscaled inputs in terms of increasing training stability, a further experiment is conducted on the Beijing land-use data set. A comparison is made between training the original-scale images and their downscaled versions by using the very basic FCN model. Fig. 13 shows the behavior of the training loss versus iterations and the OA on testing set versus epochs. One can observe that at the end of the training process, the oscillations of the loss curve greatly reduce when using downscaled inputs, whereas there is no significant change through iterations when using original-scale images. Additionally, the OA on the test set obtained by using downscaled input is significantly higher than that yielded on the original-scale images. To quantify the difference, the variance values of training loss are given in Table IV. The comparison validates our assumption that the use of downscaled training samples greatly increases the training stability and improves the convergence of the learning.

C. Analysis of Results: Effect of Local Feature Fusion

In the second training stage of the proposed architecture, features learned from downscaled images are fused with extracted local features to make predictions. To test the effect of the second training stage, we make a comparison of the

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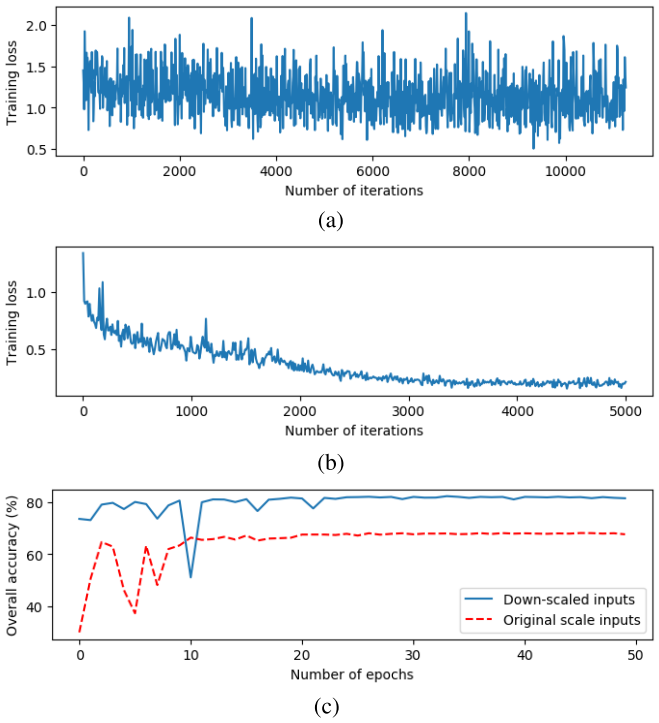


Fig. 13. Comparison of the behaviors of training loss and OA (Beijing land-use data set). (a) Training loss versus iterations (original-scale inputs). (b) Training loss versus iterations (downscaled inputs). (c) OA on the testing set versus epochs.

TABLE IV

COMPARISON OF THE VARIANCES OF THE TRAINING LOSS ON THE ORIGINAL-SCALE AND DOWNSCALED IMAGES (BEIJING LAND-USE DATA SET)

Input Images	Variance		
	All iterations	First 100 iterations	Last 100 iterations
Original-scale	0.2625	0.2464	0.2484
Down-scaled	0.1819	0.1435	**0.0210**

TABLE V

COMPARISON OF RESULTS BEFORE AND AFTER THE FUSION OF LOCAL FEATURES (ISPRS POTSDAM DATA SET)

Potsdam	Per-class F1 Score					avg.F1	OA
	Imp.surface	Building	low veg.	Tree	Car		
Before	91.47	96.35	84.65	84.98	86.40	88.77	88.85
After	92.91	97.13	87.03	87.26	95.16	91.90	90.64

results achieved: 1) before the second training stage (the prediction maps are upsampled to the original resolution to evaluate the performance) and 2) after the proposed two-stage training. As shown in Tables V and VI, the second training stage improved the OA of around 1.8% for the ISPRS Potsdam data set and of 0.4% for Beijing land-use data set. Figs. 14 and 15 show some examples of the predictions made before and after the local feature fusion on the two data sets. Although the majority of areas remain with the same predictions, the areas near object boundaries are smoothed and optimized.

TABLE VI

COMPARISON OF RESULTS BEFORE AND AFTER THE FUSION OF LOCAL FEATURES (BEIJING LAND-USE DATA SET)

Method	Per-class F1 Score						avg.F1	OA
	Cons.	Tree	Water	Farmland	Grass	Bare Soil		
Before	90.76	86.06	73.90	71.48	31.31	30.20	63.66	82.30
After	90.73	86.25	73.72	71.75	31.06	30.07	63.93	82.71

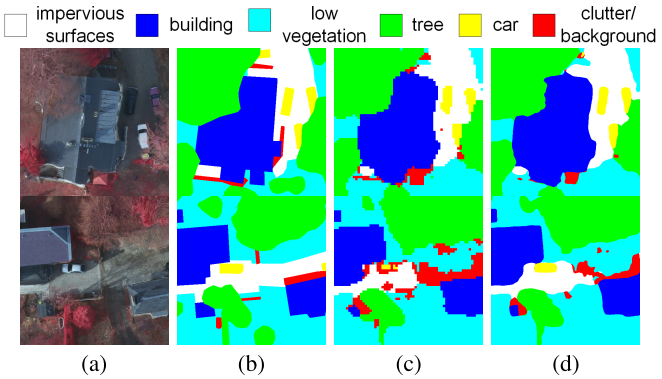


Fig. 14. Comparison of image patch before and after the fusion of local features (ISPRS Potsdam data set). (a) Image patch, (b) reference map, (c) before fusion, (d) after fusion.

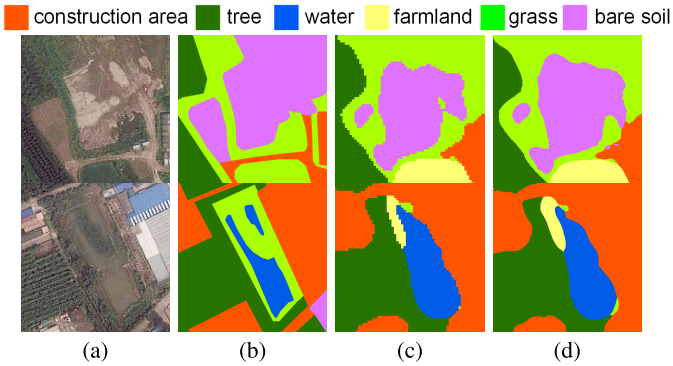


Fig. 15. Comparison of image patch before and after the fusion of local features (Beijing land-use data set). (a) Image patch, (b) reference map, (c) before fusion, (d) after fusion.

VI. CONCLUSION

In this article, a two-stage multiscale training architecture for the semantic segmentation of large-size VHR RSIs has been proposed. In the first training stage, downscaled images are used to enlarge the VRF to exploit more context information and improve the tolerance to inaccurate reference data. In the second training, local features are introduced to adjust the predictions in object boundaries. Experimental results on the ISPRS Potsdam data set and the Beijing Land-use data set show that our approach outperforms other local-patch-based training models in terms of both accuracy and stability. One of the limitations of the proposed approach is that it requires huge size training data since the downscaled training stage greatly decreases the amount of training area. Therefore, this architecture should be used for the semantic segmentation of large VHR RSIs. Additionally, ground objects detected in RSIs often appear as regions with certain shape patterns (buildings are rectangular, roads are extending lines, and so on.).

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Since the convolutional kernels used in DCNN models cannot model the direction of object contours, the shape of ground objects and some other potential geometric features are not fully exploited in existing DCNN based approaches. This is a problem that requires further studies in our future work.

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